

Original(default) inputs + Modified R-2.8

```
landerson_assign02.py U X
Assign02 > landerson_assign02.py > PredatoryCreditCard > make_payment
48 class CreditCard:
97     def _set_balance(self, balance):
98         """
99         C-2.30 At the close of Section 2.4.1, we suggest a model in which the CreditCard
100             class supports a nonpublic method, set balance(b), that could be used
101             by subclasses to affect a change to the balance, without directly accessing
102             the balance data member. Implement such a model, revising both the
103             CreditCard and PredatoryCreditCard classes accordingly
104         """
105         self._balance = balance
106
107     def charge(self, purchase):
108         """
109         A modification to charge method that would also decrement the limit
110
111         R-2.5 Use the techniques of Section 1.7 to revise the charge and make payment
112             methods of the CreditCard class to ensure that the caller sends a number
113             as a parameter.
114
115 PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
116
117 customer: John Bowman
118 bank: California Savings
119 account: 5391 0375 9387 5309
120 limit: 2500
121 balance: 2475
122 PS C:\Users\leyon\Documents\Programming\COP3410C> & C:/Users/leyon/AppData/Local/Programs/Python/Python314/python.exe c:/Users/leyon/Documents/Programmi
123
124 customer: Sally Shoo
125 bank: Vells
126 account: 1234 5678 9012 3456
127 limit: 5000
128 balance: 0
129 APR: 0.0825
130
131 visa charged $200: True
132
133 customer: Sally Shoo
134 bank: Vells
135 account: 1234 5678 9012 3456
136 limit: 5000
137 balance: 115.76221618250233
138 APR: 0.0825
139
140 visa charged $5000: False
141
142 customer: Sally Shoo
143 bank: Vells
144 account: 1234 5678 9012 3456
145 limit: 5000
146 balance: 136.66204364464738
147 APR: 0.0825
148
149 =====Following card tried to go over thier limit=====
150
151 customer: John Bowman
152 bank: California Savings
153 account: 5391 0375 9387 5309
154 limit: 2500
155 balance: 2475
156 PS C:\Users\leyon\Documents\Programming\COP3410C> 
```

- Test visa1, visa2 as predatory card.
- Test visa3 as regular.

Visa1 also tests the overdraft fee.

- Deferent test case for R-2.8

```
landerson_assign02.py U X
Assign02 > landerson_assign02.py > ...
1
2  """
3  Coded by: Leyon Anderson
4  2/14/2026
5  Assignment #2
6
7  COP3410 - Classes and OOP

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\leyon\Documents\Programming\COP3410C> & C:/Users/leyon/AppData/Local/Programs/Python/Python314/python.exe c:/Users/leyon/Documents/Programmin

customer: Sally Whoo
bank: Vells
account: 1234 5678 9012 3456
limit: 1000
balance: 0
APR: 0.0825

customer: Mary Jane
bank: BNC
account: 1234 5678 9012 2255
limit: 2000
balance: 0
APR: 0.0425

customer: Max McMill
bank: Rhase
account: 1234 5678 9000 0056
limit: 2500
balance: 500

visa1 charged $200: True
visa2 charged $250: True
visa3 charged $100: True
All visa's make a $200 payment.
Visa 1 and 2 process thier month. (Visa 3 was not predatory)

customer: Sally Whoo
bank: Vells
account: 1234 5678 9012 3456
limit: 1000
balance: 0
APR: 0.0825

customer: Mary Jane
bank: BNC
account: 1234 5678 9012 2255
limit: 2000
balance: 65.22584217522734
APR: 0.0425

customer: Max McMill
bank: Rhase
account: 1234 5678 9000 0056
limit: 2500
balance: 400

visa1 charged $5000: False

customer: Sally Whoo
bank: Vells
account: 1234 5678 9012 3456
limit: 1000
balance: 5
APR: 0.0825

=====Following card tried to go over thier limit=====

customer: John Bowman
bank: California Federal
account: 3485 0399 3395 1954
limit: 3500
balance: 0
```

Here we test for C-2.28 assuming 10 prior charges (on line #166)

```
landerson_assign02.py U X
Assign02 > landerson_assign02.py > PredatoryCreditCard > _init_
48 class CreditCard:
123     def make_payment(self,payment):
140         else:
141             self._set_balance(self.get_balance() - payment)
142
143     def __str__(self):
144         """ Returns a string representation of self """
145         return "\ncustomer: " + str(self._customer)+ "\nbank: " + str(self._bank) + "\naccount: " + str(self._account) + "\nlimit: " + str(self._limit) + "\nbalance: " + str(self._balance)
146
147     ##### Inheritance #####
148 class PredatoryCreditCard(CreditCard):    #a child of the credit card class, it is inheriting all the methods and it's allowed to use all the instances from Credit Card
149     ''' An extension to CreditCard that compounds interest and fees '''
150
151     def __init__(self, customer, bank, acct, limit, apr):
152         '''
153         Create a new predatory credit card instance.
154
155         The initial balance is zero.
156
157         customer the name of the customer (e.g., John Bowman )
158         bank the name of the bank (e.g., California Savings )
159         acct the account identifier (e.g., 5391 0375 9387 5309 )
160         limit credit limit (measured in dollars)
161         apr annual percentage rate (e.g., 0.0825 for 8.25% APR)
162         '''
163         #CreditCard.__init__(self,customer, bank, acct, limit) #use this technique or the one below
164         super().__init__(customer, bank, acct, limit) # call super constructor, this is the CreditCard initializer
165         self._apr = apr
166         self._nTrans = 10 # number of monthly charges
167         self._amtDue = {"Payments": 0, "Amt": 0} # Payment tracker
168
169     def get_apr(self):
170         ''' Returns the APR on a creditcard '''
171         return self._apr
172
173
=====Following card tried to go over thier limit=====
customer: John Bowman
bank: California Federal
account: 3485 0399 3395 1954
limit: 3500
balance: 0
PS C:\Users\leyon\Documents\Programming\COP3418C> C:/Users/leyon/AppData/Local/Programs/Python/Python314/python.exe c:/Users/leyon/Documents/Programming/COP3418C/Assign02/landerson_assign02.py
customer: Sally Whoo
bank: Vells
account: 1234 5678 9012 3456
limit: 1000
balance: 0
APR: 0.0825
visa1 charged $200: (assumes 10 prior charges) True
visa1 charged $50: True
customer: Sally Whoo
bank: Vells
account: 1234 5678 9012 3456
limit: 1000
balance: 252
APR: 0.0825
=====Following card tried to go over thier limit=====
customer: John Bowman
bank: California Federal
account: 3485 0399 3395 1954
limit: 3500
balance: 0
PS C:\Users\leyon\Documents\Programming\COP3418C> 
```